

Electro-optic non-reciprocal polarization rotation in lithium niobate

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ABSTRACT

Polarization is a fundamental degree of freedom for light and is widely leveraged in free space and fiber optics. Non-reciprocal polarization rotation, enabled via the magneto-optic Faraday effect, has been essentially unbeatable for broadband isolators and circulators. For integrated photonics foundries, however, there is still no good path to producing low-loss magneto-optic components, which has prompted a search for alternatives that do not use polarization rotation. Moreover, magneto-optic materials tend to be highly lossy, and while large (10–100 rad/cm) polarization rotation can be achieved, the key figure of merit (rotation-per-loss) is typically <1 rad/dB. Here, we demonstrate that broadband non-reciprocal polarization rotation can be produced using electro-optics in nanophotonic devices. Our demonstration leverages electro-optic inter-polarization scattering around 780 nm in lithium niobate, in which the reciprocity is broken with the help of a radiofrequency stimulus that carries synthetic momentum. While the demonstrated electro-optic polarization rotation rate is ≈ 1 rad/cm, the exceptionally low loss of lithium niobate enables non-reciprocal polarization rotators with figures of merit that are 1–2 orders of magnitude better than what is possible with magneto-optics. This approach can be replicated on III–V platforms, paving the way for high-performance lasers with co-integrated monolithic non-reciprocal devices.

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I. INTRODUCTION

The magneto-optic Faraday rotation effect (MOFE) is the foundation for all commercially fielded isolator and circulator devices available today. Its primary action is to induce directional non-reciprocity in the polarization degree of freedom for photons,^{1–3} which, when combined with polarization filters, can produce a direction dependent power-transmission function.^{4–6} The key feature that enables the dominance of the MOFE in non-reciprocal technologies is the very wide operating bandwidth over which it acts and the strength of the polarization rotation (expressed as the Verdet constant in rad/T cm) in magneto-optic materials, e.g., terbium gallium garnet (TGG) being a prominent example. For these reasons, MOFE-based non-reciprocal devices are found in all complex optical systems.

MOFE-based nonreciprocal devices do come with limitations, however, as they require specialized materials and magnetic biasing, neither of which are typically available on integrated photonics

platforms. The effect also tends to be strongly chromatic, which forces changes to material selection and adjustment of the magnetic bias in order to tune the operational wavelength. Finally, magneto-optic materials also tend to be quite lossy, with typical propagation losses approaching 50–70 dB/cm.^{7,8} This compels designers to only use the minimum amount of material and requires careful management between the strength of the polarization non-reciprocity generated (rad/cm) and the signal attenuation incurred (dB/cm), with the central figure of merit being their ratio (rad/dB). While a number of successful attempts have been made to produce on-chip MOFE devices,^{7–14} the common theme is to use the minimum allowable magneto-optic material to minimize the accompanying losses.

Because of these limitations, multiple alternative approaches for the production of non-reciprocal devices have been explored that would improve foundry compatibility, rely on lithographical patterning for wavelength tuning rather than material changes, and avoid magnetic fields for sensitive applications. The most notable

alternatives leverage spatiotemporal modulations or momentum biasing through optomechanical interactions,¹⁵ acousto-optics,^{16–22} and electro-optics^{23–28} (for a recently detailed comparison, we refer the reader to the Supplementary Information of Ref. 22). Even so, the interaction that generates the non-reciprocity is usually weak, and resonant structures are often used to enhance the non-reciprocal effect but inadvertently limit the bandwidth. Magneto-optics remain unbeatable on the bandwidth metric.

In this work, we show that a MOFE-like nonreciprocal polarization rotation effect can be achieved on-chip using electro-optic materials without the use of magneto-optics. Electro-optic materials such as lithium niobate (LiNbO₃) have been rising to prominence due to their wide band gaps, extremely low loss,^{22,29} and the possibility of producing active devices.^{22,30–33} Indeed, these materials can be turned into dynamic polarization rotators via external perturbations.^{34–36} Unlike the MOFE, however, the physics of the electro-optic polarization rotation effect is reciprocal and cannot be used directly to produce non-reciprocal devices. We show here that this reciprocity issue is resolvable by introducing a large synthetic momentum bias into the electro-optic modulation. As with previous approaches that use synthetic momenta,^{19,23,25} here we demonstrate that this approach produces a very strong non-reciprocity in the polarization conversion. Importantly, since high quality electro-optic materials can exhibit ultra-low propagation loss (<0.1 dB/cm),^{22,29} we show that the figure of merit for electro-optic non-reciprocal polarization rotators can be 1–2 orders of magnitude greater than what is possible with the best MOFE-based on-chip devices to date.^{7,8}

II. PRINCIPLE

The principle of our broadband nonreciprocal polarization rotator is described in Fig. 1(d). We consider an optical waveguide that supports two modes of types TE₀₀ and TM₀₀, i.e., with mutually transverse polarization. These guided modes are orthogonal and have distinct dispersions that are separated in momentum-frequency space by a non-zero momentum. We can induce coupling between these modes by introducing a spatial perturbation that shifts the momentum for guided photons, such as a specialized grating, though the resulting polarization rotation is completely reciprocal.^{37,38} Non-reciprocal behavior can now be introduced by adding a time-dependent component to the perturbation, that is, by using a traveling wave modulation that shifts both frequency and momentum for light.^{16,22,39–41} With this type of perturbation applied, light injected into either the TM₀₀ or TE₀₀ mode can only be converted into the transverse polarization partner mode as long as the phase matching relations hold true, that is both frequency and momentum matching conditions are satisfied (see supplementary material Secs. S1 and S2). In the specific example shown in Fig. 1(a), we apply a traveling wave modulation at frequency Ω with a forward-directed momentum such that it only permits conversion from TM polarization to TE for forward-propagating light via anti-Stokes scattering from input frequency ω_f to $\omega_f + \Omega$. Conversely, forward-propagating light in the TE polarization can only be converted to TM via Stokes scattering from $\omega_f + \Omega$ down to ω_f . On the other hand, for backward-propagating light, the phase matching is satisfied differently and the frequency shift during scattering is

reversed, such that conversion from TM to TE polarization can only take place between frequencies $\omega_b + \Omega$ (for TM) and ω_b (for TE). As a general principle, this asymmetric frequency shift effect is broadband, i.e., it works for any ω_f and ω_b and extends over a frequency range where the group velocities of the optical modes are matched.

In practice, however, these group velocities are rarely matched due to the dispersive nature of the material and the waveguide geometry. In order to understand the implications, we consider the anti-Stokes process in Fig. 1(a), although identical dynamics are produced for the Stokes process. For light injected into the TM₀₀ mode in the forward direction, the anti-Stokes polarization conversion is maximized at ω_f , where perfect phase matching occurs, and as a function of detuning, the conversion exhibits a sinc² profile that originates due to phase accumulation [Fig. 1(b)]. A detailed discussion is provided in the supplementary material, Sec. S2. On the other hand, due to the mismatched dispersion, light entering the TE₀₀ mode in the backward direction can only optimally undergo anti-Stokes scattering back into the TM₀₀ mode at a different frequency ω_b . This produces a direction-dependent spectral shift in the sinc² profile [Fig. 1(b)] and results in a nonreciprocal conversion process for the forward and backward directions. The spectral shift can be analytically evaluated as $\omega_f - \omega_b = \Omega (n_g^{\text{TM}} + n_g^{\text{TE}}) / (n_g^{\text{TM}} - n_g^{\text{TE}})$ and increases with larger values of the applied modulation frequency Ω and with better matching of the group indices. In fact, if the group indices are exactly matched, then the phase matching in the opposite direction is never satisfied, and the bandwidth becomes infinite in theory with limits imposed by other practical constraints. In the cases where this shift $\omega_f - \omega_b$ is well resolved, i.e., $(n_g^{\text{TM}} + n_g^{\text{TE}}) \Omega \mathcal{L} / \pi c \gg 1$ (see the supplementary material, Sec. S3), where $\mathcal{L}$ is the total interaction length, the device can exhibit large mode conversion in the forward direction but negligible conversion in the backward direction over some frequency bands [Fig. 1(d)]. Furthermore, this non-reciprocal behavior is maximized if the spectral shift aligns the peak of the sinc² in one direction with a null in the opposite direction of propagation. The above-mentioned analyses are discussed in detail in the supplementary material Sec. S3.

In addition to the above-mentioned phase matching requirements, we must ensure a non-zero overlap integral between the transverse polarization modes and the traveling wave modulation that provides the rotation action. Such inter-polarization scattering can be achieved electro-optically,^{34,42,43} i.e., using second order optical nonlinearity, which produces refractive index change under an applied radio-frequency (RF) electric field. The overlap integral requirement can then be summarized as $\int \sum_{ijk} \mathcal{E}_i^{\text{TM}}(\mathbf{r}_\perp) \mathcal{E}_j^{\text{TE}}(\mathbf{r}_\perp) r_{ijk} \mathcal{E}_k^{\text{RF}}(\mathbf{r}_\perp) d\mathbf{r}_\perp \neq 0$, where $\mathcal{E}^{\text{TE}}(\mathbf{r}_\perp)$, $\mathcal{E}^{\text{TM}}(\mathbf{r}_\perp)$, and $\mathcal{E}^{\text{RF}}(\mathbf{r}_\perp)$ are the transverse field distributions of optical and applied RF modes, respectively (see the supplementary material, Sec. S1), and the subscripts (*ijk*) refer to coordinate axes for the material's electro-optic tensor. This requirement is visually depicted in Fig. 1(c). The overlap integral can be interpreted as a specific requirement of non-zero electro-optic coefficients r_{ijk} for $i \neq j$,^{44,45} which are significant in many electro-optic materials, such as LiNbO₃,^{34,42,43} GaAs,^{46,47} InP,⁴⁷ and InGaP.⁴⁸ These materials, therefore, give us a path to realize non-reciprocal polarization rotation in foundry compatible processes without requiring

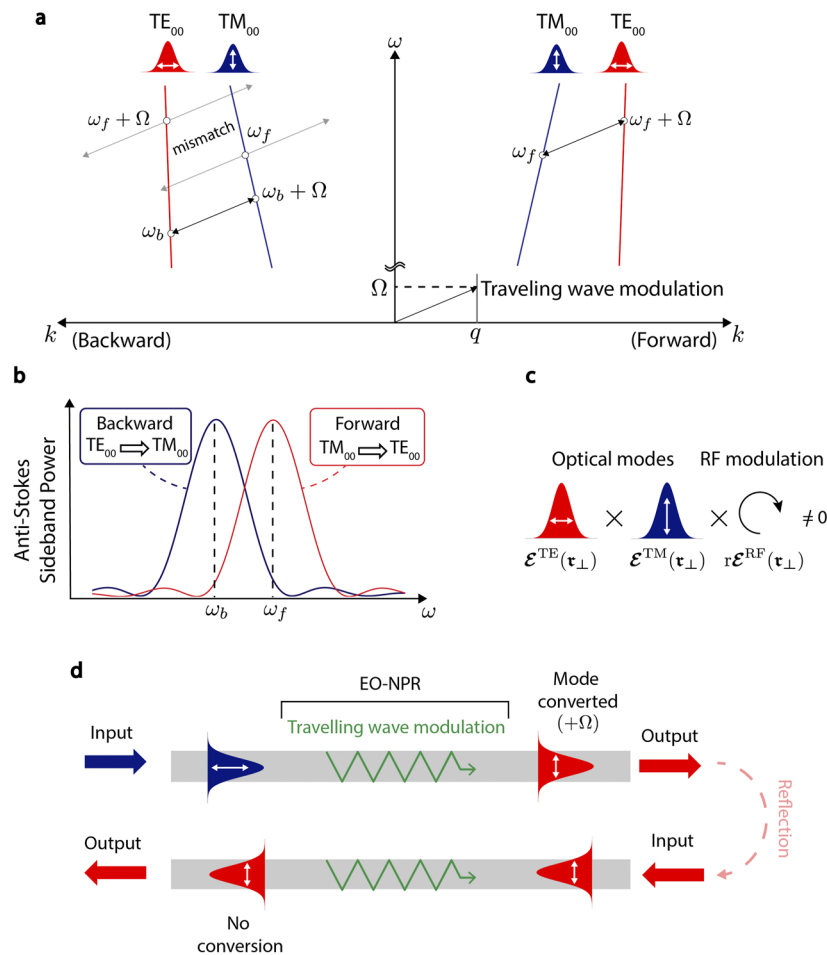


FIG. 1. Principle of non-reciprocal polarization rotation via traveling wave induced interband scattering. (a) We consider here the fundamental TE_{00} and TM_{00} modes of a waveguide with linear dispersion. A spatial modulation with large momentum applied to the waveguide has the ability to scatter photons between the modes and achieve polarization rotation. If this modulation additionally has a time-domain component, i.e., results in a traveling wave with frequency Ω and momentum q , the phase-matching is only satisfied for one direction of propagation at a selected optical probing frequency. (b) The group velocities for the modes are not matched in the general case, which results in a sinc^2 spectral dependence of the scattering process and a propagation direction-dependent spectral shift in the scattering profiles. Here, we illustrate the anti-Stokes case only. (c) The applied modulation must utilize suitable electro-optic coefficients r that break the orthogonality of the polarization. (d) An illustrative example of the resulting behavior of the electro-optic non-reciprocal polarization rotator (EO-NPR) is presented.

magneto-optics by just applying appropriately designed RF stimuli. As an important observation, the best chip-scale magnetless optical isolators to date have used materials that are not easy to co-integrate with active III–V photonics, and this electro-optic approach could help traverse that barrier.

In order to bridge the momentum gap between the optical modes, as described earlier, the spatiotemporally propagating RF field profile needs to exhibit a low group velocity, that is, large momentum for low frequency. This is relatively simple if using acoustic phonons, as in past studies,^{16–18,20–22} as they can carry significant momentum. Unfortunately, RF modes are not capable of providing sufficient momentum for modest frequencies.^{30,33,49} To circumvent this issue, we generate synthetic momentum for the RF excitation along the waveguide by engineering a split

electrode structure.^{19,23,25,50} Each electrode (of pitch L) is provided a single tone RF stimulus at frequency Ω with a relative phase shift of $\Delta\phi$ between electrodes [see Fig. 2(a)]. The dominant component of the resulting spatial phase profile produces an effective momentum at $q = \Delta\phi/L$ that can bridge the momentum gap and, in principle, could be tuned dynamically with a suitably adaptive design. In addition, this synthetic momentum can be made relatively large in order to compensate for the fabrication imperfections that can hinder the phase matching condition. A detailed discussion of this synthetic momentum, along with its primary and higher order Fourier components, is provided in the supplementary material, Sec. 4. For this work, we employed a three-phase periodicity to produce the required synthetic momentum.

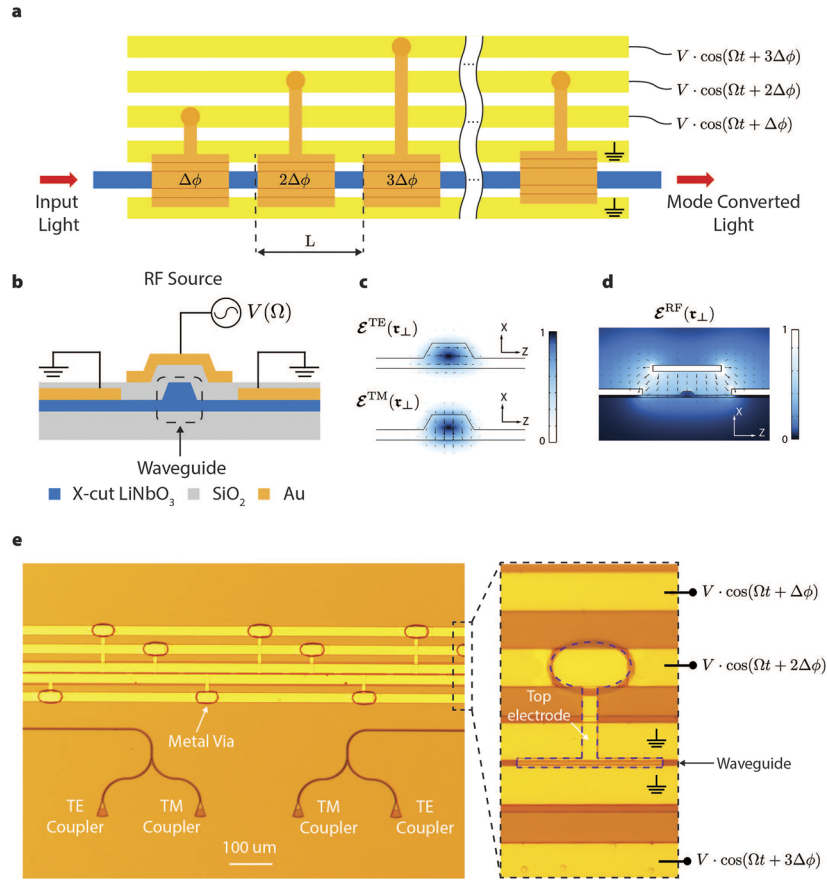


FIG. 2. Implementation of a 780 nm electro-optic non-reciprocal polarization rotator (EO-NPR) in lithium niobate. (a) A plan view of the EO-NPR device. The traveling wave modulation is realized by using a split electrode structure driven by three RF bus lines. The momentum is set by the relative phase shift $\Delta\phi$, irrespective of the frequency Ω . The incoming photons within the waveguide (blue) are polarization-converted via the electro-optic effect, which is realized by the top (dark yellow) and bottom (light yellow) electrodes. (b) Cross-sectional schematic of the device showing the electrodes (yellow), the lithium niobate (LN) waveguide (blue), and the oxide cladding (grey). The ground electrodes and the RF stimulus are also labeled to show the RF mode that is excited. (c) Finite element simulation of the TE_{00} and TM_{00} optical modes supported by the waveguide. (d) Simulated RF mode shows the vertically symmetric electric field profile that is required to realize polarization rotation. (e) True-color microscope image of the device. The LN nano-photonic structures appear in dark orange, and the gold electrodes appear in yellow. The inset shows a zoomed-in view of a single electrode.

III. DEVICE DESIGN AND FABRICATION

For experimental demonstration, we developed an electro-optic non-reciprocal polarization rotator (EO-NPR) near 780 nm using a 300 nm X-cut LiNbO_3 on insulator (LNOI) integrated photonics platform. Lithium niobate (LN) is an ideal material for these experiments since its wide bandgap provides a broad transparency window spanning 250–5300 nm that enables integrated nanophotonics with low propagation loss (<0.1 dB/cm)^{22,29} and its second-order optical non-linearities enable very efficient electro-optic modulation.⁴⁴ A plan view of the EO-NPR device is presented in Fig. 2(a), and a cross-section of the waveguide is presented in Fig. 2(b). The waveguide was designed to support only TE_{00} and TM_{00} modes near 780 nm, and details on the dimensions are provided in the supplementary material, Sec. S6. We oriented the waveguide along the Y-crystal direction (X–Z plane) to harness the r_{42} coefficient of LN (the double index notation is explained in the supplementary material, Sec. S1).

At both ends of the waveguide, we fabricated 50:50 Y-splitters followed by grating couplers that are customized to the TE and TM modes near 780 nm for selecting the polarization. Electro-optic modulation is enabled through a stimulus electrode above the waveguide, with ground electrodes on either side of the waveguide. The optical modes supported by the waveguide are presented in Fig. 2(c). We also optimized the electrode distance to mitigate any excessive optical loss while maintaining a large electric field for electro-optic modulation (see the supplementary material, Sec. S6). An RF voltage applied on the top electrode generates the symmetric vertical field profile shown in Fig. 2(d), which is required to activate the r_{42} coefficient of LiNbO_3 . This configuration also conveniently produces a mirror symmetric horizontal electric field, which is important to suppress unwanted intramodal scattering due to the large r_{33} coefficient of LiNbO_3 . The top electrodes are driven using three bus lines that provide the input voltages $V \cdot \cos(\Omega t + n\Delta\phi)$ at a frequency of Ω with relative phase $\Delta\phi = 2\pi/3$ for $n = 1, 2, 3$, which

ensures a periodic spatial phase. Figure 2(e) presents a microscope view of a portion of the device showing dedicated grating couplers used for selectively injecting TE and TM light and the electrode configuration used for generating the synthetic RF traveling wave. The electrode pitch was set to $L = 120 \mu\text{m}$ with a total length of the interaction region $\mathcal{L} = 1.08 \text{ mm}$ (three modulation periods were implemented).

IV. RESULTS

We first characterize the primary radio-frequency (RF) and optical components of the EO-NPR. The RF electrodes are individually characterized by an S-parameter measurement with a vector network analyzer. The RF reflection spectrum (see the supplementary material, Sec. S5) shows an RC response with no resonances, which confirms that the excitation of each electrode is spatially uniform and non-propagating for the designed geometry. This ensures that the split electrode structure will produce the desired momentum bias. The typical insertion loss of the TE_{00} and TM_{00} gratings is identified by measuring the corresponding modal transmission through the waveguide with no stimulus applied, and we estimate the typical per-grating loss to be 6 and 11 dB for the TE_{00} and TM_{00} modes, respectively. Later, we use these measurements in the supplementary material, Sec. S7, to calibrate the experimental results.

In order to measure the electro-optic polarization rotation effect, we used an optical heterodyne measurement system (Fig. 3, details in Sec. S7), which enables the separate characterization of input optical probes (i.e., carriers) and their sidebands. The polarization selective grating couplers additionally provide filtering action and assist with the resolution of the inter-polarization scattering rate. We normalize all measurements with respect to the injected carrier power. The inter-polarization scattering experiment

is depicted in Fig. 4(a). An optical carrier signal ω_p is injected into the waveguide TM_{00} mode via port 1, while the output light is monitored on the TE_{00} output at port 2. A synthetic traveling RF wave is generated by applying $\approx 7.1 \text{ V}$ (sinusoidal amplitude) to each electrode at 3 GHz with a few different selections of relative phase $\Delta\phi$, as described below. The adjustment of this relative phase changes the momentum bias for a given frequency modulation and alters the modulation process.

First, to observe the spectral dependence of this process, which results from the group index difference between the modes, we measure the generated sidebands for different wavelengths [Fig. 4(b)]. For this demonstration, we selected the phase to be positive ($\Delta\phi = 2\pi/3$) so that the RF stimulus wave travels in the same direction as the optical carrier signal. This results in strong anti-Stokes scattering into the TE_{00} mode since there is no density of states available for Stokes generation [Fig. 4(a)]. In Fig. 4(b), we see that the anti-Stokes signal is strongly produced around 780 nm, and the modulation efficiency drops as we move away from this wavelength following the expected sinc^2 shape with a $\Delta\lambda_{\text{BW}} \approx 2.42 \text{ nm}$, which is in agreement with the theory (see the supplementary material, Sec. S2). We also observe that the Stokes sideband is 25 dB lower, which confirms that the RF input produces an inter-polarization scattering process. Calibrations for these measurements are discussed in the supplementary material, Sec. S7.

To further demonstrate the effect of the synthetic momentum bias, we also tested our devices with $\Delta\phi = -2\pi/3$, which reverses the direction of the RF stimulus. As an aside, this is equivalent to injecting the TE carrier signal from Port 2 while keeping the phase at $\Delta\phi = 2\pi/3$. The modulation was measured near the strongest point (around 780 nm) and compared to the $\Delta\phi = 2\pi/3$ case, with results summarized in Fig. 4(c). We observe that the $\Delta\phi = -2\pi/3$ case flips the single sideband generation strongly toward the Stokes sideband, as expected. This demonstrates an asymmetry in single sideband generation due to the directionality of the traveling wave and

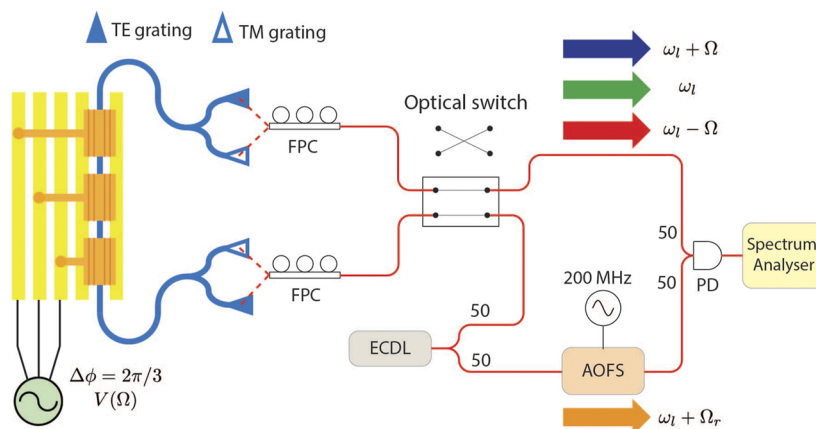


FIG. 3. Experimental measurement setup. Light (770–780 nm) from a fiber-coupled external cavity diode laser (ECDL) is split into a probe path and a reference path with a 50:50 ratio. Fiber polarization controllers (FPCs) are used to adjust the light coupled into the chip prior to the grating couplers. An optical switch is used to change the measurement direction through the device. In the reference path, we introduce an acousto-optic frequency shifter (AOFS) to offset the frequency by 200 MHz, which enables heterodyne detection through a high-speed photodetector (PD) when the probe and reference paths are recombined. The on-chip electrodes are modulated using synchronized RF signal generators with a fixed relative phase offset.

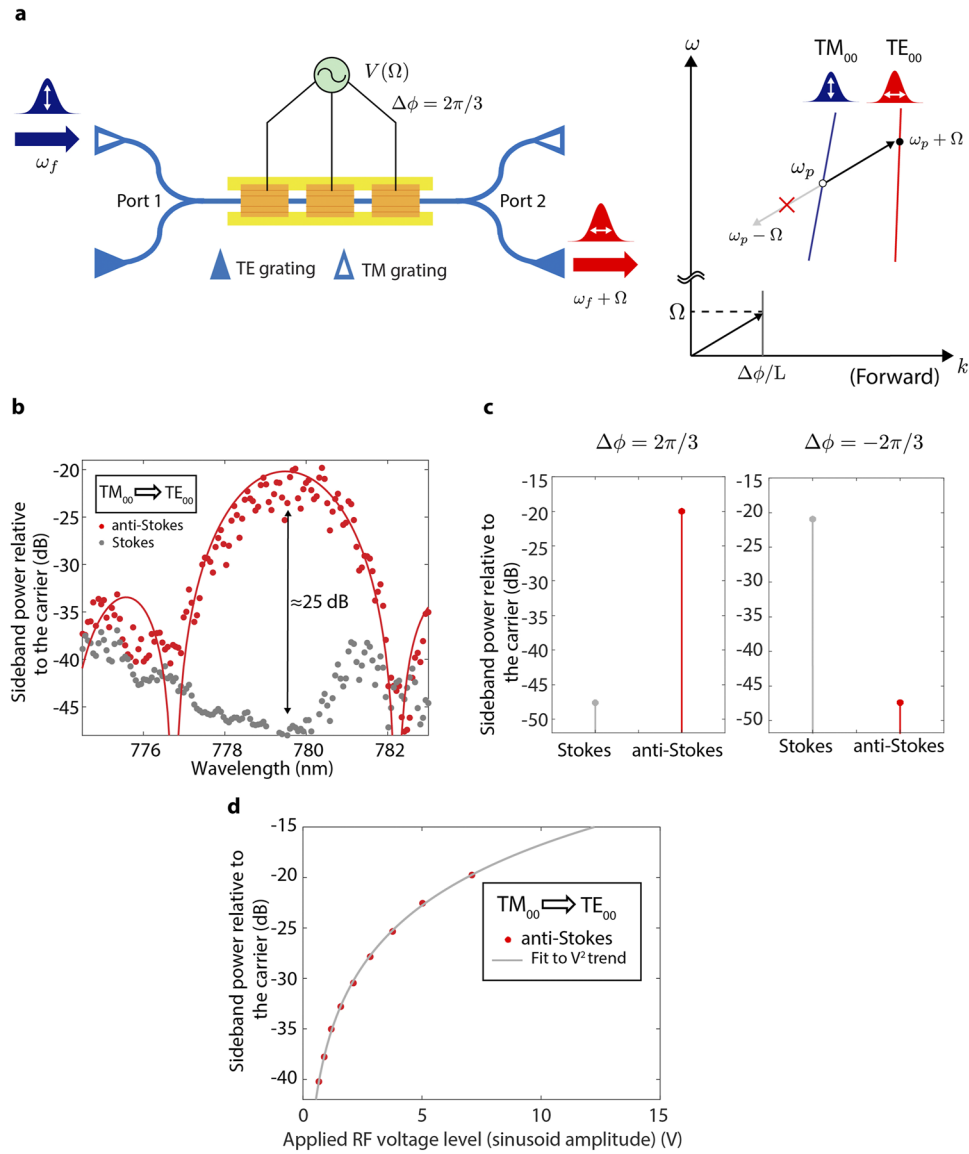


FIG. 4. Experimental demonstration of single sideband inter-polarization scattering and broken time-reversal symmetry. (a) An RF traveling wave is applied to the EO-NPR via electrodes having three-phase periodicity, which generates the required momentum ($\Delta\phi/L$). For $\Delta\phi = 2\pi/3$, only anti-Stokes scattering is generated, as illustrated in the band diagram, since there are no available states for Stokes scattering. (b) With the RF stimulus set at $\Omega = 2\pi \times 3$ GHz, we measure the modulation efficiency in the TE₀₀ mode as a function of the input wavelength. Since the phase matching condition changes with frequency, the modulation response displays a sinc^2 spectral characteristic (the solid line is fit with theory, presented in supplementary material, Sec. S2) for anti-Stokes scattering. This experiment confirms that the time-reversal symmetry is broken over a wide bandwidth and is clearly measurable over at least 5 nm. Data normalization is discussed in the supplementary material, Sec. S7. (c) Measured peak electro-optic inter-polarization scattering efficiency (occurs around 780 nm) with applied $\Delta\phi = \pm 2\pi/3$ confirms the asymmetric scattering and the breaking of time-reversal symmetry. (d) We plot the measured sideband power (relative to the carrier) for different applied voltage levels, which shows a quadratic dependence ($\propto V^2$) as expected from the theory.

confirms that the applied RF field breaks the time reversal symmetry of the optical waveguide, leading to direction dependent mode conversion and frequency shift for the input light. We also tested this effect at different RF excitation voltage levels, which we plot for the anti-Stokes sideband ($\Delta\phi = 2\pi/3$ case) in Fig. 4(d). We confirm that the sideband power (relative to the carrier) has a quadratic

dependence on the applied voltage ($\propto V^2$), as is expected from any electro-optic modulator. At the maximum voltage level, we observe $\approx 1.04\%$ inter-polarization conversion over the 1.08 mm interaction length, corresponding to a polarization rotation rate of 0.94 rad/cm (see also the supplementary material, Sec. S7). Similar to the Verdet constant that quantifies the magneto-optic polarization

rotation normalized to the applied axial magnetic field, here we can divide by the applied voltage ≈ 7.1 V (sinusoid amplitude) to obtain a normalized figure of merit ≈ 0.13 rad/V cm.

From this result, we additionally estimate the half-wave voltage product to be around 11.85 V cm, and the interaction length can be increased further to achieve full conversion (see the supplementary material, Sec. S7), even though this decreases the available RF bandwidth. In principle, as this device consumes negligible power other than parasitic resistive losses, we should be able to increase the electrode voltage with the help of RF transformers, which would in turn improve the modal conversion rate. We can also implement on-chip LC resonators on the RF bus lines, a method that is routinely employed in quantum microwave to optical conversion systems^{51,52} to help enhance the RF field amplitude. As a point of comparison, the devices that have previously demonstrated *reciprocal*

polarization rotation in Ti-diffused thin-film LNOI waveguides have shown a best case half-wave voltage product of ≈ 30 V cm^{34,35} with a similar X-cut crystal. This highlights an advantage of the thin-film LNOI approach due to enhanced optical confinement in the nanophotonic waveguide, which permits closer placement of the electrodes.

In order to compare our EO polarization rotation performance with MOFE-based counterparts, we use a well-established figure of merit (FoM)—the polarization rotation rate divided by propagation loss (i.e., rad/dB)—which is commonly used for non-reciprocal polarization rotators. Previous results from some best-in-class MOFE devices in Ref. 7 estimate a maximum FoM of 0.66 rad/dB (52.36 rad/cm with a propagation loss of 78.95 dB/cm). Similarly, for the MO devices in Ref. 8, this FoM is estimated at 0.75 rad/dB (102.97 rad/cm with a propagation loss of 137.20 dB/cm). With

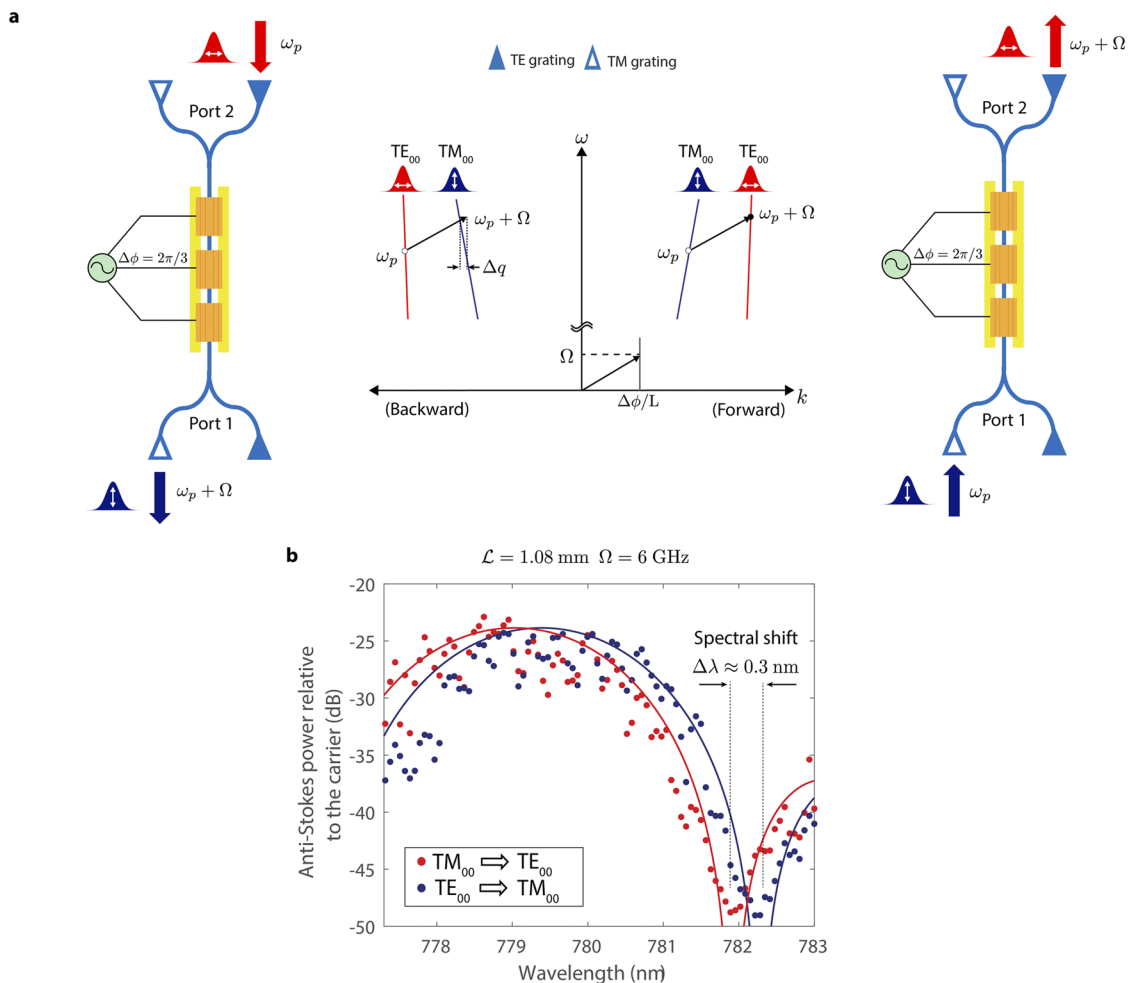


FIG. 5. Experimental demonstration of non-reciprocal polarization rotation. (a) We consider anti-Stokes scattering for a probe of frequency ω_p with a fixed $\Delta\phi = 2\pi/3$. In the forward direction, with input on TM_{00} , the phase-matching condition is satisfied for scattering into the TE_{00} mode. In the backward direction, anti-Stokes scattering is only satisfied if the input is on TE_{00} and scattered into TM_{00} . Even so, a non-reciprocal phase mismatch Δq prohibits this process from occurring at the same frequency in both directions. (b) Experimental results for a device with 1.08 mm length and a modulation frequency of 6 GHz, with ≈ 7.1 V (sinusoidal amplitude) applied to each electrode. The measurements confirm the spectrally shifted sinc² responses and demonstrate non-reciprocal conversion. Solid lines are a fit for the theory.

our current results in lithium niobate, we achieved a comparatively much lower polarization rotation rate of 0.94 rad/cm, which could be improved with a larger applied voltage. Even so, lithium niobate offers incredibly low loss with a typical loss rate of ≈ 0.1 dB/cm (our best loss rate was previously measured at around 0.02 dB/cm near 780 nm²²). Using these numbers, we can estimate a non-reciprocal polarization rotation FoM ranging between 10 and 50 rad/dB, which is 1-2 orders of magnitude greater than on-chip MO solutions.

The above-mentioned experiment establishes asymmetry in TM_{00} to TE_{00} conversion depending on the direction of the traveling wave and demonstrates broken time-reversal symmetry that is broadband and extends over at least 5 nm. However, this is not the typical mode of operation for a MOFE device. Since polarization is an important defining characteristic of what constitutes a “port,” the more relevant case is that of an undesirable reflection coming back into the output mode (here TE_{00}), as previously illustrated in Fig. 1(e). For this, we test the case illustrated in Fig. 5(a), i.e., TM_{00} to TE_{00} in one direction and TE_{00} to TM_{00} in the opposite direction, for a chosen stimulus configuration. We note here that depending on the choice of $\Delta\phi$ for the RF stimulus, this results in either only Stokes or only anti-Stokes conversion for both inputs. For our demonstration, we tested the anti-Stokes conversion case to show the non-reciprocal polarization rotation. As described earlier, the forward and backward sideband conversion processes should display a shifted sinc^2 response depending on the group index of the modes and the modulation frequency. To increase the relative shift between these sinc^2 functions and clearly show a strong non-reciprocal effect, we increased the RF frequency Ω to 6 GHz and applied the same voltage to the electrodes (≈ 7.1 V sinusoid amplitude). This flexibility of the stimulus frequency is possible since the momentum shift is set by the electrode pitch L and the relative phase shift $\Delta\phi$. The resulting experimental observations are presented in Fig. 5(b). We observe a spectral shift of 0.3 nm between the sinc^2 conversion responses, with a peak sideband non-reciprocity of ≈ 9 dB, proving the non-reciprocal polarization rotation effect. This spectral shift is also in good agreement with our

theoretical calculations (in the supplementary material, Sec. S3) and can be increased by using larger RF frequencies as described earlier. Unfortunately, it was not possible to increase the frequency beyond 6 GHz in our current devices due to bandwidth limitations introduced by the electrodes (discussed in the supplementary material, Sec. S5).

We can use the extracted data from Fig. 5(b) to extrapolate the operational characteristics of a device that could reach complete inter-polarization conversion with a longer interaction length $\mathcal{L} = 16.74$ mm (see the supplementary material, Sec. S7). This simulation is presented in Fig. 5(c), where we additionally set a higher RF frequency ($\Omega = 31.4$ GHz) so that the conversion peaks and valleys are aligned for the highest contrast. This ability to independently set the contrast and the conversion efficiency is quite advantageous from a design perspective. Such long interaction lengths can be readily obtained in smaller form factors through resonant approaches^{16,19,22} or through the wrapping of the waveguide into compact spiral structures.^{21,33,53–55} It is also possible to reduce the estimated voltage and length requirement by introducing a fourth electrode below the optical waveguide to improve the verticality of the applied RF field, as presented in supplementary material Sec. S8. Our calculations show that a further figure of merit improvement of $\approx 15\%$ is possible in such a device.

V. CONCLUSIONS AND OUTLOOK

As a key point of distinction with the traditional MOFE, the electro-optic non-reciprocal polarization rotation that we have demonstrated is always accompanied by a fixed frequency shift of Ω that is set by the RF input tone. Even so, this EO-NPR effect can be leveraged in a manner similar to the MOFE by introducing appropriate polarization selective filters and splitters. For instance, we can produce a linear optical isolator without any frequency shift by using two full-conversion EO-NPRs with an intermediate polarization selective filter [Fig. 6(b)]. As illustrated, in the forward direction, the first EO-NPR converts the TM input to a TE signal

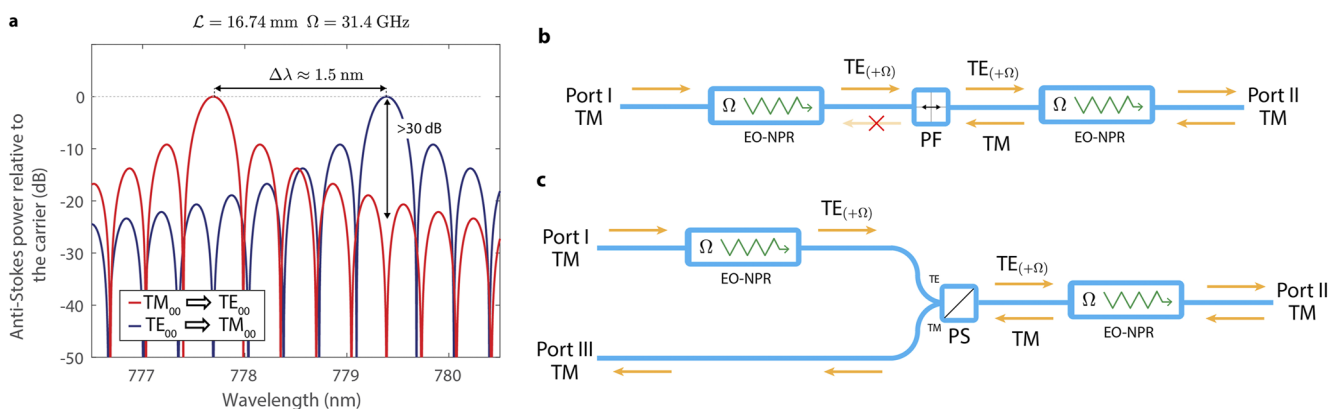


FIG. 6. Extrapolated experimental results for full inter-polarization conversion and potential designs of a 2-port isolator and 3-port circulator. (a) Extrapolating from the experimental results in Fig. 5, we can simulate a “full-conversion” EO-NPR that produces complete inter-polarization scattering by simply increasing the interaction length to $\mathcal{L} = 16.74$ mm. The applied RF frequency, which independently sets the contrast, is also increased to 31.4 GHz to optimize the alignment between peaks and valleys in opposite directions. (b) We can construct a linear non-frequency-shifting isolator by using two EO-NPRs with an intermediate polarization filter (PF). (c) A linear non-frequency shifting circulator can be implemented by replacing the PF in the isolator design with a polarization splitter (PS) that sends returning signals to a different output port.

that passes through the filter and is subsequently converted back to TM. In the reverse direction, the first EO-NPR does not perform the conversion, which allows the filter to reject the TM mode. Linear 3-port circulator operation, again without any frequency shift, can also be achieved by replacing the intermediate polarization filter in the isolator design by a polarization splitter that leads to a drop port [Fig. 6(c)]. These proposed EO-NPR based designs for isolators and circulators have similar complexity to devices that use the MOFE.

The ability to perform non-reciprocal polarization rotation has been generally absent from photonic integrated circuits, yet devices that leverage this effect are widely used in critical locations within photonic systems. Acousto-optic inter-polarization scattering has been used previously^{19,56} as an alternative approach to address this challenge; however, in practice, the power consumption can be prohibitive. This issue appears because the input RF power must be electro-mechanically transduced into an acoustic wave within the material that propagates away from the excitation electrodes and is mechanically dissipated. In this work, we have demonstrated that an ultralow-loss magnetless alternative to the MOFE can also be realized by leveraging electro-optic coefficients ($r_{ijk, i \neq j}$) that are found in many photonic materials. This list includes active III-V platforms based on GaAs,^{46,47} InP,⁴⁷ and InGaP,⁴⁸ potentially making it feasible to co-integrate lasers with high-bandwidth optical isolators and circulators without requiring heterogeneous integration. Compared to acousto-optic approaches, this electro-optic technique does not fundamentally consume any input RF power since the device electrodes represent an open-circuit load, which is an advantage that is shared by lumped electro-optic modulators. Of course there are some practical limits due to parasitic resistive losses in the materials and the signal generation circuitry. Unlike magneto-optic non-reciprocal polarization rotators, this electro-optic approach also decouples the full-conversion interaction length from contrast control (the latter can be manipulated by the applied RF frequency), which can allow dynamic adjustment even after fabrication. In addition, the operational wavelength range for EO-NPRs is set lithographically, which makes short wavelength non-reciprocal devices much more accessible, especially for atom-photon integration and low-loss quantum photonics.

SUPPLEMENTARY MATERIAL

Please refer to the supplementary material for comprehensive theoretical derivations regarding non-reciprocal electro-optic modulation, detailed device design information, and experimental procedures.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Oğulcan E. Örsel: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Resources (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Gaurav Bahl:** Funding acquisition (equal); Investigation (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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